



Live-Sound Pressure That Is Not Measured In Decibels

An investigation into live-sound culture and the interpersonal pressures that influence audio engineer choice, the live-sound “experience,” and the health and wellbeing of participants

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Abstract

Audio engineer choice is affected by the influences and pressures within the live sound culture. This study analyzes the data from a 2020 international survey of audio engineers and focuses on live sound interpersonal interactions through the lens of social responsibility. As live sound participants, audio engineers, event organizers, performers, and audiences often have differing expectations due to their knowledge, preference, bias, ego, financial investment, sense of responsibility, proclivity towards risk or safety, and more. At live sound events, the final gatekeeper between the sound source and sound reinforcement is the audio engineer, who consciously or subconsciously weighs these expectations against the pressure they feel, their own motivations, and their knowledge and ability. Within the decision-making process, an audio engineer’s performance benefits not from the absence of interpersonal pressure, but instead from an appropriate balance of influences and pressures. This balance affects audio engineer development, investment, focus, enjoyment, satisfaction, and audio mix choices, which then impacts other live sound participants’ experience, health, well-being, and event viability.

Keywords Live sound culture · Sound reinforcement · Audio engineer choice · Social responsibility · Social pressure · Eustress · Traditional expectation · Loud · Sound exposure · Artistic integrity · Audience satisfaction · Venue and event safety · Hearing management · Critical listening · Multiplicity of opinion · Shared vision · Mutual benefit · Appropriate mix

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1 Introduction

At socially responsible live sound events, the people who make decisions must clearly understand the implications of the choices that affect the experience, the health, safety, well-being of others, and the event's viability. In practice, this is not always the case. Some audio engineers, for example, are trained and well-informed on best-practice, while others do what they can with what they know. Some feel pressure to perform in ways they would rather not, while others stick to their beliefs despite external influence. While audio engineers may not have overall event responsibility, their perception of their responsibility also affects their choices. Currently, little, if any, other research has investigated the cultural pressures and influences affecting the people that ultimately control event sound reinforcement; the audio engineers.

Not every live event could consider their sound as high-quality, appropriate to the genre for most people attending, and safe. The sound levels at venues, for example, are often louder than many audiences prefer (Beach & Gilliver, 2019). Hearing impairment from high sound exposure is also more common in live sound participants than other people (McGinnity et al., 2021; Rawool, 2012). Given these findings, there must be overriding reasons within the live sound culture as to why sound levels and exposure commonly remain high. By analyzing the data from an international audio engineer survey conducted in 2020, this study explores the live sound culture and why audio engineers make the choices they do.

Corporate social responsibility (CSR) seeks an optimum situation that satisfies most people without harming others. It balances the multiplicity of interests for mutual benefit, justice, cooperation, and the common good (Garriga & Melé, 2004). This definition of CSR offers an appropriate framework to compare and assess similar aspects within the live sound culture. Where possible, this study uses the respondents' direct quotes to reduce misinterpretation and allow for ongoing discussion.

2 Literature Overview

An organizational group or team may contain subgroups and individuals. A cultural group shares a set of beliefs, behavioral norms, and behavioral expectations that is recognizably different from those of other groups (Barth, 1969). Social psychology defines a group as “two or more independent individuals who influence one another through social interactions that commonly include structures involving roles and norms, a degree of cohesiveness, and shared goals” (APA, 2023). Subgroups are “subsets of team members that are each characterized by a unique form or degree of interdependence” and interact differently with each other than with other team members by sharing a cultural value, resource, knowledge, “meaning,” or identities that is unique from other team members (Carton & Cummings, 2012; Osgerby, 2014). Given these definitions, under an overarching venue or event ‘group,’ event ‘participants’ can be regarded as a collection of subgroups made up of audio engineers, audience, management, regulatory providers, performers, and more. While sharing

commonalities in overall purpose and synergy, each may have their own unique identity, knowledge, and values, and differing beliefs, behavioral norms, and expectations.

Resources on the technical aspects of sound production are widespread. The damaging effects of high sound exposure on audio engineers and musicians are also well documented (Chasin, 2009, 2014; Compton, 2022; Di Stadio et al., 2018; McGinnity et al., 2021; Rawool, 2012). Studies have shown that the majority of patrons at loud events are dissatisfied with current sound levels, that most audiences do not demand high sound levels, and that lower levels would not reduce attendance at live sound events (Beach & Gilliver, 2019; Gilles et al., 2014; Mercier & Hohmann, 2002). Mulder et al. (2022) confirm the need for more audio engineer education and training. Although these sources cover aspects of live sound culture, they do not address the associated influences and pressures that lead audio engineers to deliver sound at times that does not optimize mutual benefit, is for the common good, or is without the potential to cause harm.

Physical and social pressures are essential within live sound events. Physically, sound pressure vibrations are formed and amplified, and affect a person or an object in some physical way. This may include activating hearing mechanisms, causing skin sensations and resonance within body cavities, moving objects like microphone diaphragms, loud-speaker elements, or causing glasses to rattle. Social pressures can be motivational, but equally affect action. For example, they may encourage further learning and development to perform a role at a high standard. Social pressures may arise from interactions between event organisers, musicians or attendees that push participants to perform differently than they would prefer, like requests to change the sound level or mix balance.

Pressure is a physical force exerted on or against an object by something in contact with it. As a verb, pressure is persuasion, coercion, intimidation, or feelings of stress caused by the need to do or make someone do something (Oxford University Press, 2001). These definitions imply that no pressure exists when there is no opposition. Indeed, in the absence of physical pressures, live sound events would not occur as sound is fluctuations in positive and negative air pressure. Understanding and working with physical sound pressure properties and within sound pressure limits enables audio engineers to maximize the creative potential of sound while minimizing undesirable distortion and tone that can cause unpleasantness, distraction, and damage (Compton, 2022; Everest, 2007).

Work pressures exist in many occupations. When work demands/pressures exceed a person's perceived abilities, occupational stress can result and, if prolonged, can lead to negative consequences for the worker and their professional efficiency (Maslach & Leiter, 2016; Westman & Eden, 1996; Yusoff et al., 2012). Although the verb definition of pressure is essentially a negative view, Westman and Eden (1996) propose that work pressure at an optimal level (eustress) can enhance performance.

In Social Responsibilities of the Businessman, Bowen suggests that social responsibility should be a basic foundation for human interactions. Social responsibility seeks an optimum situation for “the satisfaction of all people involved...or at least satisfies most [people] without being detrimental to others...balancing the burden versus benefit” (Bowen et al., 2013). Since Bowen's original 1953 publication, researchers have expanded on social responsibility concepts, many of which are analyzed, compared, and summarized by Garriga and Melé (2004), where the four generally accepted CSR theories are: Instrumental, Political, Integrative, and Ethical. These are explained further as:

- meeting objectives that produce long-term profits,
- responsibly using [business] power,
- integrating social demands, and
- contributing to a good society by doing what is ethically correct.

Further debate on the numerous texts within Garriga and Melé's (2004) summary and critique of the CSR model is beyond the scope of this paper and would add little, if any, to the core of this study. However, the commonalities between the CSR texts are sufficient for use as a measure of the under-explored pressures and influences within the live sound culture. Bowen recognized the difficulties in implementing corporate social responsibility. CSR practice, however, would see participants actively align their decision-making power to make improvements by considering their own views, the person ultimately accountable, and the community at large (Bowen et al., 2013).

As described above, the literature on the effects of loud sounds on hearing, general workplace pressures, and responsible practice are common. However, previous research has yet to explore the competing cultural pressures within live venues and events that challenge safety and responsibility ideals. The research presented in this paper through a mixed-method design is one step towards addressing that gap by integrating the quantitative research from related studies with an analysis of live sound event culture, mainly through the words of venue and event audio engineers.

3 Aims

The aim of this study is to gain an understanding of the factors that influence audio engineer's decisions within the live sound culture. As little scholarly information currently exists on the social factors, it is hoped the insights from this study will help provide a foundation for future research on the field, inform practice, and guide future directions for education, standards, and audio engineering practice.

The choices audio engineers make can enhance or reduce audience well-being, a concept that is within the domain of positive psychology. In addition, gaining insights into the culture of the workforce is related to positive psychology in terms of positive organisational science.

4 Method

Like many people, audio engineers continue with an activity if they find the rewards and benefits outweigh the costs and negative aspects. By investigating what audio engineers most and least enjoy, and their personal experiences within the culture, this study aims to achieve insight into matters that affect them. During the survey design process there was no known existing survey for audio engineers to answer the research aims, so the author collaborated with other researchers to create the survey used in this study.

The survey was informed by a review of the existing literature, and covered topics relating to audio engineering choices that were identified within the literature, as well as many missing from it. The topics in the survey included demographic information

(geographic location, age, gender, education/training, work experience), information about how audio engineers learn their skills, personal skills, how much they enjoy their work, how satisfied they are, what factors relate to their satisfaction, what pressure and responsibilities they feel, and what factors impact their sound levels during events. These topics relate to the study aims by providing information about the audio engineering culture. Due to this study's exploratory nature, we aimed only to describe the audio engineering culture rather than answer research hypotheses.

This study used a convergent mixed-methods design, as described by Clark (2019). In this method, the quantitative and qualitative analyses are merged to provide an integrated analysis drawing interpretations based on the combined strengths of both data sets. According to Clark, this type of analysis is holistic and can be used to gain insight into a topic, so it was an appropriate approach to apply to an investigation of audio engineering culture. While conducting this survey, a different set of researchers also launched an extensive audio engineer quantitative study (Mulder et al., 2021, 2022). In contrast to Mulder et al.'s survey design, the mixed-method approach clarifies why engineers chose a particular answer and allows for unexpected responses beyond the survey creator's design.

In 2020, 203 audio engineers were invited to complete a mixed-methods online questionnaire that explored audio engineering culture and the various influences that impact audio engineer choice. The survey was built using Qualtrics and was comprised of 134 questions that include check boxes, free-text entry, forced-choice options and Likert scales. Some items were contingent on each respondent's previous answers; therefore, the sample size differed for many survey items. The survey is included in the Appendix, and shows both the questions and the 'skip and display logic.' A significant proportion of this survey was quantitative but also allowed people to expand on their responses. The survey took approximately 30–60 min to complete. Participants could start the survey, and exit, or complete at a later stage, and no feedback was given as to right or wrong answers. Many of the questions were designed in such a way that unexpected responses could be submitted.

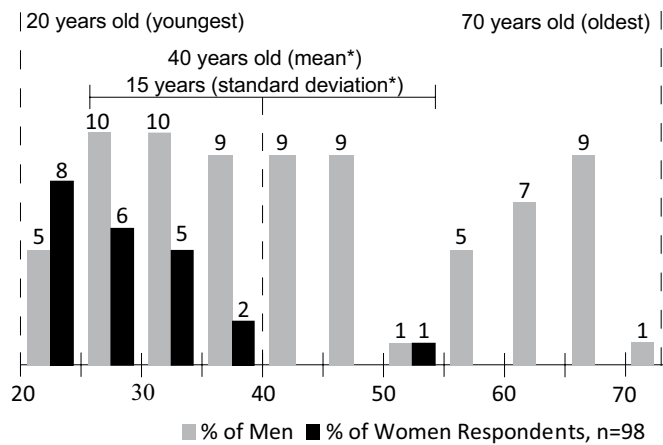
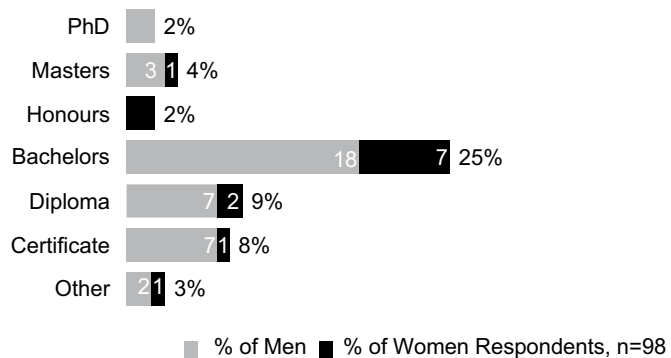
Ninety-six people completed the survey approved by the University of Canterbury Ethics Committee. Themes and patterns were drawn from their responses and coded via the Qualtrics Report function, SPSS Statistics software, and manual comparisons.

4.1 Participant Demographics

The survey participants were audio engineers from various international live sound businesses, the Audio Engineering Society, New Zealand churches, Women in Live Music, Sound Girls, and three individuals from Facebook audio engineering groups. They came from the six World Health Organization (WHO) groupings except for South East Asia, where no engineer chose to be part of this survey (Table 1). Figure 1 summarizes their age and biological sex demographic groupings.

Table 1 The location of the 2020 survey participants

Number of Participants		World Health Organization Region
Men	Women	
1	0	Africa
17	2	Americas
1	1	Eastern Mediterranean
6	16	Europe
0	0	South East Asia
49	3	Western Pacific

Fig. 1 The number of survey respondents within five-year age groupings. * refer to the age of the entire dataset of participants**Fig. 2** The highest audio qualification held by audio engineers

4.2 Education and Learning

An audio engineer's musical background, education and experience may influence the decisions they make as an engineer at a live event. Appendix Questions 4.1 – 4.9 asked respondents their musical and education background, finding that seventy-nine percent ($n=95$) of respondents play an instrument, perform on stage, compose, or arrange music. Fifty-four percent of respondents had formal audio engineer training. Figure 2 displays their highest audio qualification.

4.3 Audio Work Experience

In Figs. 3, 4, 5, 6 and 7, audio engineers describe their audio work type and frequency, experience, and average audience attendance at their events.

Fig. 3 The percentage of respondents that identified as volunteer and professional engineers

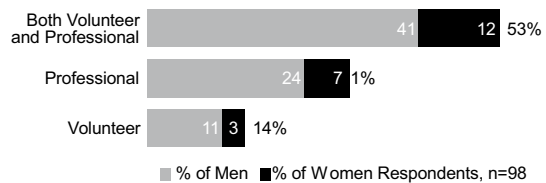


Fig. 4 The type of audio work respondents perform

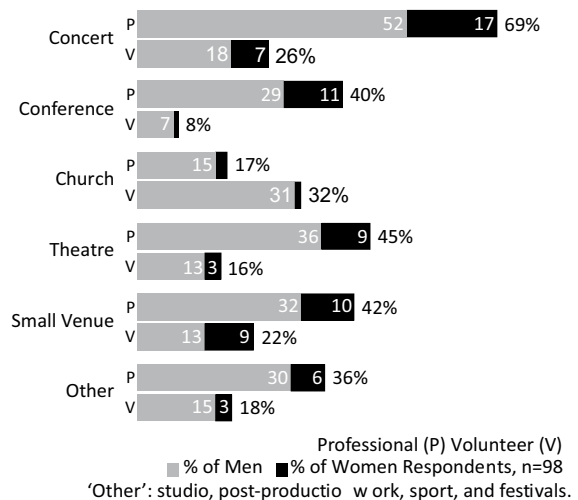


Fig. 5 The number of times audio engineers mix per month

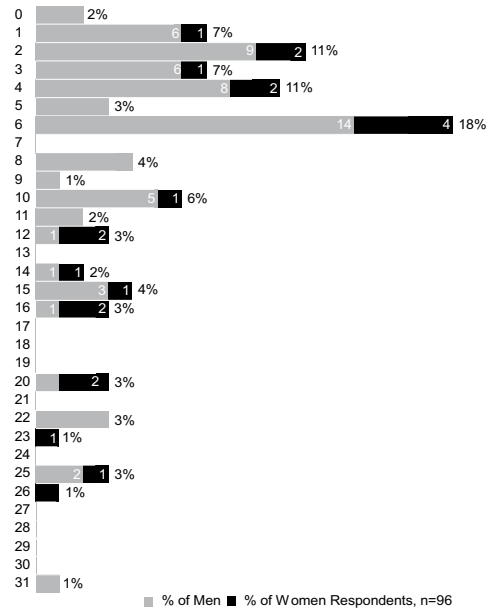


Fig. 6 Audio engineer experience (years)

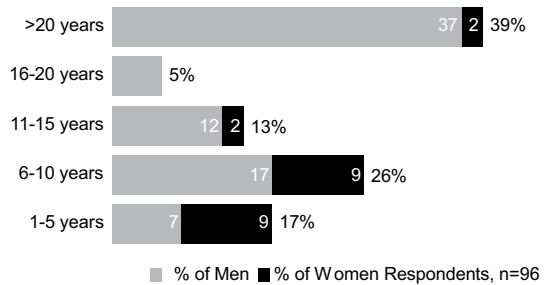
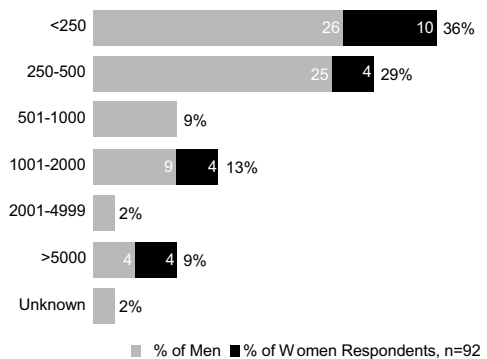


Fig. 7 The average attendance size at respondent events



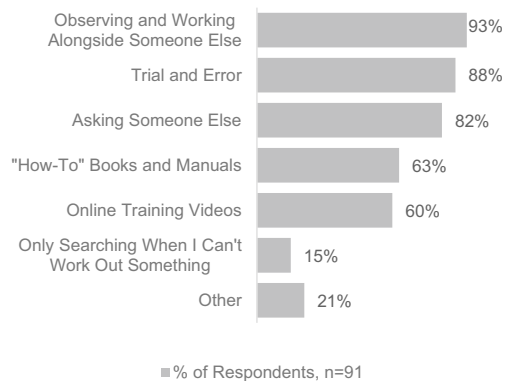
5 Results

This section uses the 2020 survey data to explore the live sound culture through the eyes of the audio engineer. The content provides insight into their ongoing learning process, job satisfaction, motivations, interpersonal influences and pressures, and more.

5.1 Education Impact

Deeper investigation into the 59% of respondents with formal audio qualifications (Fig. 2) revealed that their studies provided a practical foundation for their audio engineering skills. Responses to Survey Questions 4.4 – 4.5 found that for 19% of audio engineers, their qualifications had “*a lot*” of impact on their audio engineering abilities, while 10% and 8% reported “*some*” or “*not much*” impact, respectively. Three percent identified networking as the primary benefit. Apart from formal audio education, respondents described more than one other way they learned their skills (Fig. 8).

Fig. 8 How audio engineers otherwise learn their skills. “Other”: manufacturer workshops and informal training demonstrations



5.2 Personal Skills and Character

The survey (Appendix—Questions 4.10 – 4.11) asked audio engineers what were the most important skills and character traits that audio engineers should possess. The qualitative responses often spanned more than one theme; therefore, the total exceeds 100%. Despite arguably the strongest association with audio engineering, only 50% identified technical ability as one of the most important skills; sixty percent identified musicality. The two highest polled attributes relate to personal character (68%) and people skills (65%), many of which would be favorable within any profession. These attributes include:

- having good communication skills,
- being prepared,

- being able to focus,
- possessing stamina,
- exhibiting patience,
- being able to make good decisions under pressure,
- able to take criticism,
- being teachable,
- demonstrating humility,
- possessing a perceived calming nature,
- being open-minded,
- having a good work ethic,
- acting professionally even if they don't particularly like the material,
- being confident,
- being committed,
- being able to see the big picture,
- being able to own their mistakes,
- possessing a positive attitude,
- being passionate,
- being self-motivated, and
- being someone who shows initiative.

5.3 Enjoyment and Satisfaction

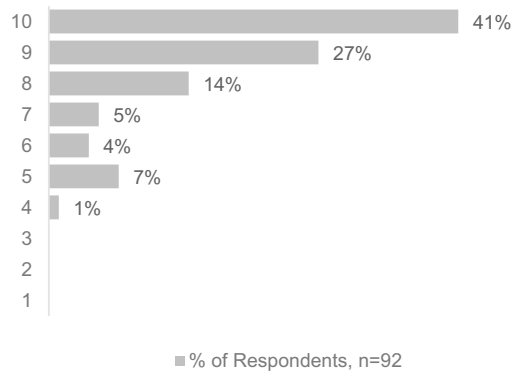
Most audio engineers enjoy and are satisfied with their role, elaborating that they like:

- the variety of work,
- the art and creativity,
- the challenge of making a great sound,
- the opportunity to support the band and enhance the emotion that musicians on stage bring to an audience,
- the thinking and ongoing learning required,
- the contact with the people involved,
- the process of improving their craft through practice,
- the role of inspiring others,
- the satisfaction and reward from getting a good sound,
- observing positive participant engagement,
- the blending of a passion for music and technology, and
- travel.

Respondents rated their audio engineering enjoyment and satisfaction (Appendix—Questions 5.1 – 5.5) out of ten on a Likert scale with ten being the highest level of satisfaction and enjoyment, and one, the least (Fig. 9). Sixty-eight percent of respondents placed their satisfaction at the highest levels.

Several audio engineers commented:

Fig. 9 How audio engineers enjoy their work (scale of 1 – 10)



“I can do a 70-hour week and still be hungry for more gigs at the end.”

“I have the best job in the world! I still can’t believe I get paid to travel and go to concerts all the time.”

“I like playing a large role in ‘the sound,’ not just what we hear...I like getting it right.”

“I enjoy the methodicalness of set up and pack down; the solving of problems, seeing how things work; and being able to serve.”

“I like the responsibility of it...to provide the best experience possible. I like being able to craft a sound to sound exactly how I want it.”

“It’s a sickness, perhaps. But I love making performances or programs sound great, be accessible to the audience (without straining), and having an artistic hand in how the delivered result sounds.”

“I do not feel that I’m working doing this!”

“Very satisfying when all elements come together to deliver the required outcome.”

“I get a buzz every time I turn the console on, and get the pleasure of mixing bands.”

“I really enjoy mixing when the band is good, the gear is good and there are no issues...on those days I would score a 10.”

“It’s a very weird, challenging, rewarding occupation.”

“It’s such a pleasure...on lucky days to enhance the emotion that musicians on stage bring to an audience.”

“Because I love it! Even on bad days!”

5.3.1 A Mix of the Good and the Bad

The process and practice of mixing audio to enhance and reinforce a performer’s sound were predominantly recognized as positive experiences. However, the role does have challenges that potentially create pressure. Engineers said:

“I really really love what I do, but it’s tough on the mind, body and general life outside work.”

“Yes, there can be struggles but I believe they are there to challenge you and to develop your abilities.”

“...stressful when I do mix a larger event, as it requires more focus.”

“... it’s a bit ungrateful and the hours are sometimes rough.”

“Some days I love it, some days it’s just stressful. I think when the artists are giving you request after request and you just can’t keep up...[and] also the stress of bigger crowds knowing something could crap out and all eyes will be on you.”

“Some days, it’s brilliant. Some days it’s an endurance test.”

“...depends on the genre I’m working with...some music is soul-destroying.”

“It’s not a career you will ever survive in if you rate it any lower than a 10 as it is a passion beyond ‘just a job’.”

With many variables affecting the resulting audio, the role may not suit perfectionists. Two respondents wrote:

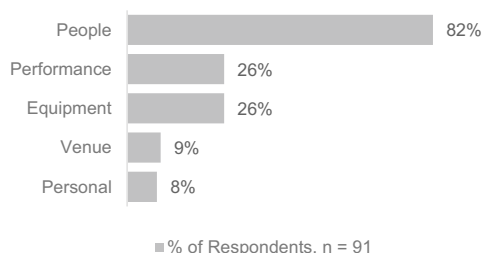
“I would say one show out of every 100 I am almost completely satisfied with. It is a stressful occupation. I am also a perfectionist, which means I am usually left slightly disappointed.”

“I’m...constantly afraid of messing up.”

5.3.2 Factors that Provide the Least Enjoyment and Satisfaction

Understanding the factors that provide the least amount of enjoyment and satisfaction to an audio engineer, may reveal hidden components that impact the overall performance. Responses to survey question 5.6 show that eighty-two percent of audio engineers regard the interactions with other people as providing the least satisfaction and enjoyment in the audio engineer role (Fig. 10). The choice of performance content and delivery, equipment, venue (and mix position) followed in ranking as contributing to the next least enjoyable aspects. Expanding on these factors (that aren’t people-related) is outside the scope of this paper. Their mention, however, is relevant as someone has chosen each option. These choices then affect the audio engineer’s ability to optimize the sound. The personal skills (8%) mentioned in Fig. 10 include knowledge and experience, confidence, critical listening, fatigue, health, etc.

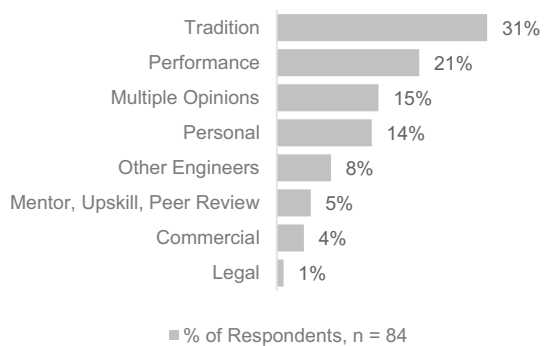
Fig. 10 Factors that audio engineers least enjoy



5.4 Pressure in the Live Sound Industry?

As with many activities, performing at a high standard under time and financial constraints can generate pressure for those involved. While time and financial pressures should be anticipated and expected, over 71% of audio engineers (Appendix—Questions 9.6 and 9.7) said they experience pressure to perform audio tasks differently than they would otherwise prefer (Fig. 11). Sections 5.4.1–5.4.6 expands on the themes outlined in Fig. 11.

Fig. 11 The pressures on audio engineers to deliver audio differently than they would otherwise prefer



5.4.1 Tradition

Tradition and expectations from the band, the audience, and the event organisers, about how live events have been carried out before (Appendix—Questions 9.1—9.2), can guide each towards continuing a known formula that has proved successful in the past. Respondents said:

“The main purpose of a lot of engineers is to please a production with the standard they expect.”

“You deliver a product that they and their audience expect.”

“Some engineers are constrained by the traditional approach to that genre which they are expected to produce.”

“If certain music is always mixed in a certain way, people have come to expect that, so it’s very common to go along with that, instead of against it.”

5.4.2 Performance Pressure (Event Organizer / Artist / Manager)

A number of aspects influence an audio engineers mix. In Survey Questions 9.1, 9.2, and 9.6 – 9.8 we asked respondents to consider the following factors on their live-event performance:

Content: musicality that follows genre, band catalog, emotional connection, and the type of event.

Technical: the audio engineer considerations of the physics of sound involving: clarity, blend, energy, feedback, meters, ‘noncontrollable’ noise, SPL legal requirements, and venue acoustics.

Engineer: Engineer expectations and preferences.

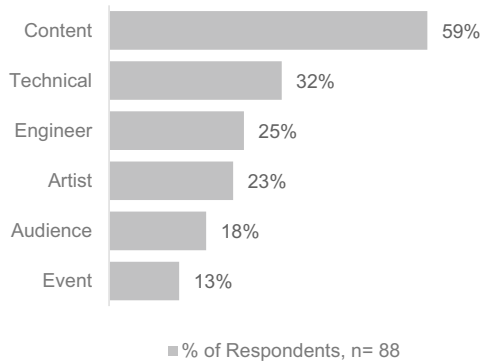
Artist: Artist expectations and preferences.

Audience: Audience expectations and preferences.

Event: The venue, mix position, event organizer rules, expectations, and staff/event management.

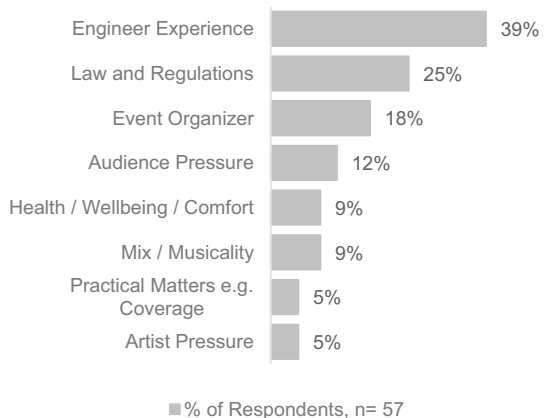
Audio engineers evaluated the influences on their mix decisions and, thereby, the overall sound. Figure 12 is a summary of these influences:

Fig. 12 How audio engineers determine their overall sound



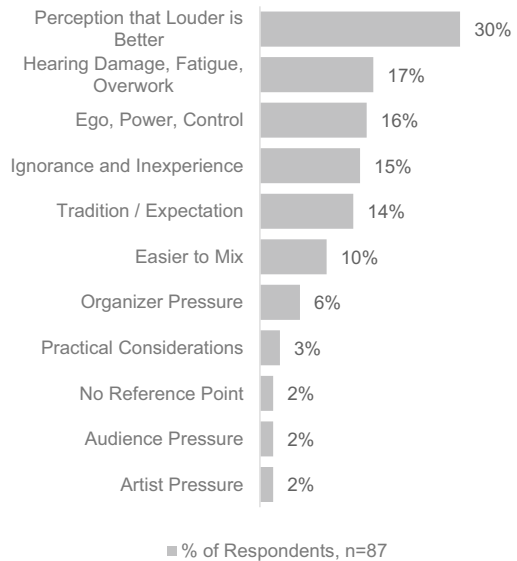
With regard to event loudness and creating a favorable and acceptable experience, some respondents reported that people had differing expectations on how loud an event should be. Survey Questions 6.1 – 6.8, 10.1–10.3 asked about the factors influencing the audio engineer’s sound-level decision making process (Fig. 13) with ‘Engineer Experience’ and meeting the ‘Law and Regulations’ forming the basis for most (64%) of the respondents’ decisions.

Fig. 13 The factors that determine event SPL target/limits



While external influences/pressures exist, audio engineers predominantly determine sound levels (Fig. 13). Some engineers choose to mix at higher levels than others. Respondents suggested reasons for this, answering Survey Question 9.8 with the “perception that louder is better” as the most cited reason (Fig. 14).

Fig. 14 Why some audio engineers choose to “mix” at higher sound levels than others



During a live event, a producer or manager may push for a particular level or ‘sound’ (Appendix—Questions 6.5 – 6.6):

“I can’t say how many times I have had a manager say to [make] it louder to grab people’s attention.”

“The producer may have a differing idea as to what they expect from their event and as a professional, it is your job to see that idea through to the best of your ability.”

“Shows are generally loud. I feel there is an underlying pressure to match this.”

“There is definitely a push for music to be louder.”

The cultural and traditional etiquette for festival-type events with multiple artists and engineers can also influence mix and sound level decisions:

“In a festival, if your band is not the first one in the setlist, you have to adjust to the level/tone/etc of the prior band, at least in the beginning.”

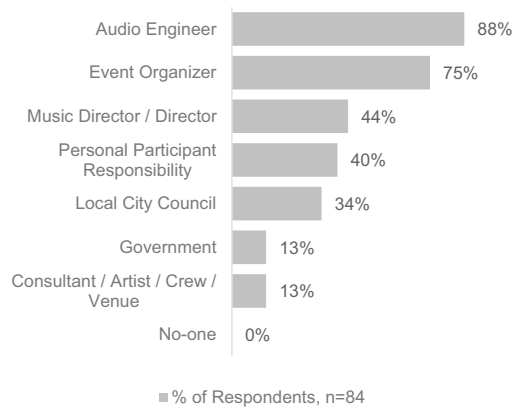
“A lot of people believe they have to mix at least as loud as the person/band before them to make an impact.”

“In festival situations, many bands think louder means better, so there is pressure on the engineer to be louder than everyone before them.”

Noting how the ‘push for louder music’ affects sound quality, 10% of respondents said that louder music was easier to mix. In contrast, others describe the ‘loudness wars’ where poorly executed sound production can result in narrower dynamics, damaged

transients, and sonic distortion. Although the engineer makes the final decision on mix and sound levels, the results from Survey Questions 10.1 – 10.3 show in Fig. 15 who audio engineers believe is responsible for managing safe noise levels/exposure at live events. Although event organizers have direct responsibility for an event, 88% of audio engineers still believe they are responsible for managing safe noise exposure (Fig. 15).

Fig. 15 Who audio engineers believe is responsible for managing safe noise exposure at live events



Only 4% of respondents to Survey Questions 9.1 said that the organizer’s commercial/financial considerations affected their mix decisions (Fig. 11). As shown in Fig. 10, 82% reported interpersonal and organizational matters negatively affected their enjoyment and satisfaction (Appendix—Question 5.6), specifically mentioning: poor leadership or organization, miscommunication, criticism, ingratitude and disrespect, and poor attitudes by artists, techs, organizers, or audiences. One commented that “...some participants are employed with not enough ability to fulfill the expectations of the role.” Four respondents mentioned “...being confronted with sexism at work.” One said this occurred “...daily.” Others noted not enough “...preparation time,” “...last-minute changes,” and “...unrealistic expectations without the budget to realize them.” Some pointed out that those running events commonly “...had a lack of understanding [about] what is involved or required...” to get a good sound. One said organizers had exhibited “...a lack of creativity (art).” Another said:

“I often find the attitude and behavior of some selected engineers, promoters, and managers in this sector to be completely unacceptable. These individuals undermine working environments and generally ruin what should otherwise be an experience where all technical staff are working toward the collective goal – successful[ly] delivering the show.”

Respondents expressed the expectation that live performances sound “... like the record,” which has influenced production, requiring engineers to adapt to using more technology and prerecorded tracks to supplement and enhance stage content. Pressure to use more complex tools puts less-experienced engineers in front of equipment they may not fully understand. Eighty percent of audio engineers (Appendix—Questions 4.12 – 4.14) believe that the number of tool/process options

affects some engineers adversely. By prioritizing the tools and processes, some engineers lose focus on the final product or can become overwhelmed with the complexity or number of available choices:

“...driving engineers to use all the tools even if they don’t need to (a perception that complexity is better).”

“...also, there will always be certain dynamic/FX units/plugin that are ‘en vogue’ at the time, and people will use them maybe without really understanding what they’re doing.”

“Marketing within the industry convinces many engineers that they need all the latest effects/plugin to do their work. This shifts the focus from the stage to the mixing desk, usually with poor results.”

5.4.3 The Differing Opinions of Live sound Participants

One respondent summed up the underlying sentiment on differing opinions within live sound events saying that for different people, the sound might be both:

“...too quiet and too loud at the same time. Too much reverb, or not enough.”

From Survey Question 9.2, several engineers elaborated on these differences of opinion and expectation:

“Audio is largely an art therefore heavily opinionated on what is the ‘right option’.”

“The bigger the show, the more interests are involved.”

“Everyone judges everyone else’s mix, regardless of if they say anything. Everyone usually has an opinion that they could do it better or different.”

“There is always pressure because everyone wants something different but it is the engineer’s job to do what they think is best.”

“Usually, there is a team consensus on what is required from those on stage, but you, after all, are there to serve, and it is a matter of balancing the needs of the musicians and audience. Many have unrealistic expectations... however, if it makes sense and they have the authority, I will try and accommodate their wishes without a fuss.”

“We work for the artist. Ultimately our job is to do what they want. Aside from the artist, there is the manager, agent, promoter, significant others of the artist, and friends, and everyone has an opinion. The drummer’s wife thinks the drummer should be the loudest thing in the mix, and so does every drummer in the audience. The guitar player’s wife feels the same about the guitar, etc.”

5.4.4 Personal

An engineer’s personality can affect their ability to process the various influences, pressures, and expectations. While Section 5.2 lists many of these positive character traits, three respondents summed up the essential skills they believe audio engineers should possess (Appendix – Question 4.10):

“Ability to focus and discriminate. Good hearing. Willingness to stick with things to figure them out. Sense of musicality - what things should sound like. Logical mind that can develop repeatable habits/steps to success. Ability to fake being calm, even if you aren’t. Ability to visualize the signal flow. Ability to hear oddities and make good guesses about where they are coming from and how to solve them (clipping, slapback, distortion, bad [signal to noise], impedance mismatch, etc. An ‘always scanning’ mind that can be one with the music but at the same time able to notice when a driver has blown, or a coat has been draped over a speaker cabinet.”

“I think there is a range of skills that are important. Technical proficiency... communication and teamwork. A familiarity with the programme material: how it has been presented in the past, and what the typical expectations are. Self-initiative/motivation...an ability to pay attention to detail while maintaining a view of the ‘bigger picture’.”

“I would divide these skills into two categories: skills that make you good at your craft and skills that make you successful in the music business. For the former (craft), I would say first and foremost is your ability to listen critically, followed by an understanding of the music you’re working on, and operate/manipulate the tools needed to achieve musical goals. The business side of things is very different: humility, interpersonal communication skills, the ability to balance a budget, and tenacity in pursuing work.”

The novice or less-experienced engineer may face more uncertainty or make less-than-ideal choices due to their inexperience:

“It is sometimes difficult, especially when starting out, to be strong in your opinion when you are responsible for a mix and have a lot of other strong opinions on what you should do.”

“I think for newer people, it’s harder to stand on what you think as you may not be sure of yourself, etc.”

“...because I know sometimes, I draw a bad mix, and I don’t know everything about sound.”

“...many people will make what they like the reference point to what sounds good, instead of mixing in the style traditional to the genre.”

Critical listening is an essential audio engineer’s tool (Compton, 2022). Any impact on the ability of this tool to function increases the variables involved in achieving a favorable mix. Aside from physical hearing damage, 17% of respondents noted that hearing health, fatigue, and overwork affected their judgment (Fig. 14). Seven percent of those surveyed recognized the personal, physical, and mental impact of long or loud events. One said:

“If the shift is too long. Being in a loud environment more than a few hours is conducive to hearing fatigue.”

5.4.5 Other Engineers

Of all live sound participants, audio engineer peers should understand better than non-engineers the complexity of the audio engineer role and the various demands they face. Despite this presumption, results from Survey Question 9.2 show that not all respondents regard other engineers' influence as constructive:

"The industry is flooded with novices with over-exaggerated egos who don't accept constructive criticism, as well as proven industry professionals that only care about self-promotion & selling the next greatest device whilst not mentoring any of the younger kids who have a real interest in audio. The 'middle-class' of audio engineering is vanishing."

"...more about showing off than good sound."

"Each engineer is different, and in live audio, there is a big culture of trying to prove skill levels and industry knowledge by telling other people what to do."

"...the pressure ...in our industry, which is noxious, is the "macho" way of thinking about our work: who can be the loudest, have the greatest PA, etc."

"I believe it is driven by insecurity."

Some engineers did provide constructive influence:

"I always get soundies asking me what I am doing and why."

"...peer review and mentorship where lessons are passed on."

"...when you're pushed to use a particular effect or sound...it forces us to stay open to what we think is 'the right sound'."

"...we tend to copy the behavior of others whom we feel are successful, or at least working with well-known artists."

In less-than-ideal 'live' conditions, whether due to poor acoustics, poor mix position, or excess stage noise, 36% of audio engineers communicate with other technical crew, colleagues, or musicians to get advice to then change what they can.

5.4.6 Audience

By observing audience reactions, audio engineers can get a sense of and gauge the quality and appropriateness of their mix (Appendix—Question 5.5):

"I like...how I can influence how people feel."

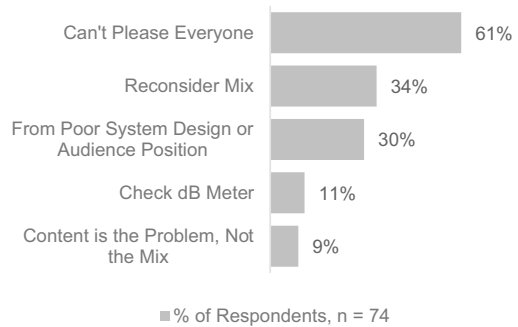
"...to witness the presence and reactions of the audience."

"It's great to get a good mix and see the resulting level of enjoyment of the people."

"Satisfying when an audience erupts into applause."

The appropriateness of a chosen sound level will relate to the event type and the participants' aims, expectations, and preferences. Seventy-eight percent of audio engineers have had an event attendee say the event audio is *'too loud.'* From Survey Questions 6.1 – 6.3, respondents described their interpretation of, or possible action resulting from, the *'too loud'* statement (Fig. 16).

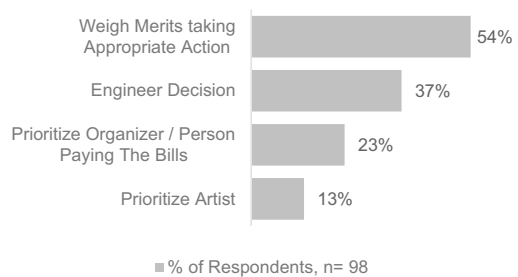
Fig. 16 How audio engineers interpret audience comments that the sound is “too loud”



5.5 How Audio Engineers Respond to Differing Opinions

As mentioned in Section 5.4.3, whether sound level-related or a combination of physics, art, tradition, bias, preference, and expectation, audio engineers must assess differing opinions. Even with a considered approach, their resulting audio mix actions are unlikely to please everybody (Appendix—Question 6.4). Figure 17 indicates engineers' responses to differing views.

Fig. 17 How audio engineers respond to differing opinions and expectations



When asked how they dealt with differing participant opinions, engineers said:

“Diplomacy is one of the greatest skills an engineer should develop. Sometimes it can be a bit of a balancing act. It mostly depends on who you are working for.”

“Always take on board criticism from those who understand or know what they want, especially those who are paying the bill. I am definitely guilty of ignoring the “I used to be a sound man” punter who had a garage set up in the '80s.”

“Do what I think sounds best, and make small tweaks if I agree with any opinions that have been offered to me. If I don't agree, I won't change it. I'll alter monitor mixes for performers.”

“Everybody is to be listened to at the right time: Performer in the rehearsal, soundcheck, pre-production, organizers at soundcheck, maybe in pre-production. [and] attendees after the concert. During the concert nobody should give an opinion, it's just about me and the music.”

“You can’t make everybody happy, so I would opt to make my employer happy.”

“The performer is always for me the most important opinion for me, within reason. As it is their emotional content. Organizers and attendees chose a performer to be there, with their whole vision of a performance.”

“You have to do what you think is best - I’m the one with the audio engineering training, and therefore I ultimately make the final call. I’m good at my job, so I trust my own ears.”

“Smile and nod.”

5.6 Potential solutions to manage each pressure

In finding ways to manage the various venue and event expectations and risks, Survey Question 14.8 asked respondents whether audio engineers that operate over $L_{Aeq}85$ dB should be required to participate in an education or certification program to improve audio engineer knowledge (Fig. 18).

Fig. 18 The % of audio engineers that believe those that operate over $L_{Aeq}85$ dB require safety training/certification



6 Discussion

Most audio engineers ‘loved’ their role for the creativity, art, challenge, variety of events, learning, and using technology to craft sound and solve problems. They did note numerous interpersonal pressures and influences that potentially affect their performance, enjoyment, and job satisfaction. In the following discussion section, these interpersonal pressures and influences have been weighed against the Corporate Social Responsibility (CSR) model that ‘seeks an optimum situation that satisfies most people without harming others, balancing the multiplicity of interests for mutual benefit, justice, cooperation, and the common good’ (Garriga & Melé, 2004).

6.1 Corporate Social Responsibility Instrumental Theory

In broad terms and applicable to live sound events, CSR Instrumental Theory suggests that events should be financially sound and benefit each stakeholder (Bowen et al., 2013). Investing well in those within their management sphere directly benefits those concerned and indirectly benefits the business’ financial success (Hillman & Keim, 2001). Any investment, therefore, must be economically beneficial

and requires each contributing participant do all they can to make the current event a financial success and future events a favorable prospect.

6.1.1 CSR Instrumental Theory: The Audio Engineer

Only 4% of audio engineers reported that commercial or financial considerations directly affected their audio engineering choices. Despite not being a primary motivation, other considerations that draw audio engineers to perform at their best will indirectly affect an event's commercial viability. We can assume that the more skills audio engineers have in the areas described in Section 5.1, and 2.3.4, the more equipped they are to assess the multiplicity of opinions and improve the likelihood of event success. We can also assume that when audio engineers perform an appropriate and high-quality audio mix, the likelihood of an event's economic success is also improved. Just over half of the respondents have had formal audio training; most are musicians or involved in music-related activities; and most improved their skills using videos, printed material, watching others, and more. Despite any training, some describe uncertainty and ignorance affecting their decision-making, and bravado leading some to act as though they knew more than they did.

Doing 'all they [audio engineers] can do to make the current and future events favorable' requires audio engineers to not only continue learning, but also to interact with other people. Ninety-three percent of respondents noted that guidance from someone else helped develop their abilities. Interactions with performers helped inform and enhance the performer's work. Good relationships between employees and organizers/employers helped manage expectations. Seventy-eight percent of engineers have had to interact with someone who suggested their mix was too loud. Therefore, given these and other interactions, navigating interpersonal relationships aptly is essential for the viability and success of event production.

6.1.2 CSR Instrumental Theory: Other Participants

The event organizer weighs the benefits against the financial investment in the people involved, the equipment, the venue, etc. Investing in highly-skilled staff and excellent equipment in acoustically appropriate venues could potentially improve the likelihood of a successful 'experience' for participants (WHO, 2022). Conversely, the event may not be economically viable if there is too much investment in these areas. Accepting less-skilled or volunteer staff/performers or poorer equipment in less-than-ideal acoustic venues may make an event a financial success but may also simultaneously compromise the 'experience' and future ticket sales. In these situations, participants may need to manage their expectations.

Not all event organizers seem aware of how their decisions affect other participants and, thereby, the event's success. Some audio engineers believed some organizers did "*not know what they [were] asking for,*" and some had "*unrealistic expectations without the budget to realize them.*" Budget constraints can

introduce time pressure and limited opportunities for rest, affecting fatigue and focus. Bakker and Demerouti note that job demands contribute most to exhaustion, whereas less-than-ideal job resources contribute most to cynicism (Bakker & Demerouti, 2017). These and other organizational choices affect employee performance, health, and attitude, potentially affecting the event's production quality and viability.

Event organizers and audio engineers may continue with traditional approaches that have proved economically successful in the past. In the case of sound levels, some have equated louder events with greater 'impact' and potential for success. Many people like loud sounds and associate them with good times, and therefore expect that live events should be loud as a cultural norm (Welch & Fremaux, 2017). Some survey respondents believe traditional expectations drove sound levels and mix decisions, and some engineers prefer to go along with these expectations rather than against them. Some noted that increased sound levels made bands easier to mix and helped to cover poor musicianship. In festivals, there is a common expectation that the sound levels of bands be at least as loud as the previous band, with the headline act, the climax (WHO, 2022), being the loudest.

The engineer, at times, will set the overall level just above the acoustic level of the loudest onstage instrument – commonly the drums, which may already acoustically be above the recommended safe SPL limits (WHO, 2022). If, however, the sound level is perceived as "*too loud*," the focus is drawn onto 'the sound' more than the content, reducing the 'impact' while also increasing the potential for hearing damage and reducing satisfaction.

Even for potential safety and sound quality improvements, an organizer may consciously or subconsciously not choose to risk an event's financial success by challenging traditional methods, expectations, and cultural norms. However, the risk may be less than organizers expect, as Beach and Gilliver (2019) notes that most loud-event patrons were dissatisfied with current sound levels. Gilles et al. (2014); and Mercier and Hohmann (2002) that most audiences do not demand high sound levels and that lower levels would not reduce attendance at live sound events. Beach and Gilliver (2019) explains further: "Rather than motivating patrons to change their behavior, encouraging venues to meet their patrons' needs and preferences may be a more successful strategy. Venue operators may find that this approach has a positive impact not only on the hearing health of patrons, but also on the economic health of their venue."

To maximize the likelihood of producing an appropriate mix, an audio engineer should be fully informed on the current sound and the effects of their audio choices. Unimpaired and unimpeded hearing with attuned critical listening skills increases the likelihood that any audio problems can be heard and addressed, and helps to recognize the actions required to enhance and optimize the current sound (Compton, 2022). The hearing damage present in more than half of audio engineers must increase the potential for compromised audio choices (Compton, 2022; McGinnity et al., 2021).

6.2 CSR Political Theory and Live Sound

CSR Political Theory concerns responsible interactions and connections, power and position, and developing ‘proper’ relationships through a shared vision (Bowen et al., 2013). Keith Davis suggests that “where social power exists, social responsibility also exists” (Davis, 1967). Melé writes that CSR for the common good must be fair, respect the dignity and fundamental rights of the individual, and contribute to their well-being both in the present and the future (Bowen et al., 2013). The WHO Safe Listening Venues and Events Guidelines echo the value of a shared vision for each event through a signed agreement by organizers, engineers, crew, and musicians (WHO, 2022).

Generally, despite ‘people matters’ being one of the most challenging aspects of the audio engineer role, many respondents also said they also enjoyed their interactions with live sound people, supporting and serving the performer and enhancing their performance.

6.2.1 CSR Political Theory: The Event Organizer

As the person ultimately responsible for an event, the event organizer holds the ultimate power, position, and inherent responsibility. As implied in Sections 6.1.1 – 6.1.2, organizers commonly must rely on the specialist skills of others. Not all organizers exhibit mutually beneficial and respectful behaviors or develop a shared vision. Some respondents noted poor interpersonal industry attitudes between live sound people. Four respondents experienced sexism; one daily.

When faced with differing opinions on audio matters, 23% of audio engineers will prioritize the decision made by the employer, organizer, or person paying the bills. Even when opinions differ, some recognize it is “...*their event, and as a professional, it is your job to see that idea through to the best of your ability.*” In contrast, one respondent said that in the end, the audio engineer should “... *do what [they] think is best – [as] the one with the audio engineering training, and therefore ultimately [qualified to] make the final call.*”

6.2.2 CSR Political Theory: The Audio Engineer

As the final gatekeeper between the sound source and sound reinforcement, the audio engineer also holds significant ‘power and position’ and responsibility. They may not, however, be recognized as such by those unaware of the technical, social, physical, and mental ability required, or how their choices affect the participant experience. In Section 5.2, respondents mentioned numerous character traits and qualities they thought audio engineers should possess. Most engineers (88%) believed they were responsible for the audio mix and sound levels. Some engineers expressed their satisfaction when audiences responded well to their mix. Some described the live sound industry as “*flooded with novices*” with “*exaggerated egos,*” a culture where some try to show their “*skill levels and industry knowledge by telling other people what to do*” and is “*more about showing off than good sound.*” One respondent

noted the pressure they felt to get a mix 'right' where "...a perfectionist can usually be left slightly disappointed." Another mentioned the fear of "...messing up."

In this pivotal role, an experienced engineer may be more capable than a novice of interacting with others responsibly, balancing time-sensitive tasks, and managing equipment, musicality, and processes, while also producing an appropriate mix. They may be able to achieve a pleasing sound, fix an issue quickly, and still "...pay attention to detail while maintaining a view of the 'bigger picture'" and avoid being "...buried in the console menus over something minor while missing the most important thing on stage."

Logically, an engineer with a reputation for producing excellent audio results, possessing attuned critical listening skills, and who fosters good working relationships is likelier than a novice to be calm under pressure and elicit trust amongst live sound participants.

6.2.3 CSR Political Theory: Audio Engineer Peers

From their experience as audio engineers, peers should understand the complexity of the role more than other people and be more empathetic to the challenges they face. Many respondents had experienced positive development through peer review, mentorship, copyable practice, or encouragement to achieve the 'right sound.' However, only 5% of respondents in the workplace identified other engineers as a positive influence. Survey respondents noted that peer and industry pressures sometimes cause engineers to use tools differently, perceiving that complexity is better.

6.2.4 Political CSR Theory: Other Participants

Audio engineers noted the challenge in weighing the multiplicity of other people's strong opinions, especially if the engineer is insecure or inexperienced, and when performers make "unreasonable requests," are "obnoxious," "unreasonable," "rude," have a "bad attitude," or are "bad at playing." Audio engineers generally seem willing to address potential issues through communication. Thirty-six percent of audio engineers mentioned that they would communicate with the band or other crew members to get advice, ninety-five percent of respondents said they would challenge performers on their sound levels, tone, or safety choices. Thirteen percent believed the artist/performer held the most valid opinions on how an event should sound, although less than half of these (5%) allowed artists to determine their SPL target/limits.

The influence and pressure from others can make some people feel unsettled or unsure about which choices are most appropriate, particularly for the less-experienced novice audio engineer. These influences and pressures can then potentially negatively impact an engineer's focus and performance. In contrast, others find eustress pressure focuses them, forcing them to be more aware, perform at a high level, develop more skills, and gain greater satisfaction "...when all elements come together to deliver the required outcome."

At times, event attendees and other participants will voice their opinions to sound engineers believing they are in a position [of power] to change sound levels. Seventy-eight percent of audio engineers have had someone say that the sound is "too loud."

However, an attendee's definition of "*too loud*" may be from factors outside the engineer's control, like the participant's proximity to the PA and stage. Low-risk sound levels may still be perceived as "*too loud*" if people cannot hear the words from the stage or have trouble hearing their own voice or the voice of the person next to them. "*Too loud*" may also be a response to the performer's musicality, style, or how the content aligns with the participant's belief system, thereby making a person feel uncomfortable. After receiving such a comment, just over a third reconsider their mix/level. Almost two-thirds believe "*you can't please everyone.*"

Audience reactions play a part in how audio engineers mix, as engineers recognize their choices and actions can "...influence how people feel." Twelve percent said audience reactions determine their overall sound level, but only 2% would change the sound level due to audience pressure. While the audience reactions may be observed and indicate how the mix translates, the audience influence is considered low in the political pecking order. In response to any audience comments, an engineer may reconsider their mix and respond with "...diplomacy" or "*smile and nod.*"

6.3 CSR Integrative Theory and Live Sound

Integrative Theory within corporate social responsibility insists on improvements as a "principle of public responsibility, and when businesses perceive a gap, they should choose a response to close it". Businesses that depend on society for their existence should, in turn, consider social values and demands (Bowen et al., 2013). Improvements that align closer to participant preference and are mindful of their experience, safety, and well-being could improve an event's appeal and ticket sales (Beach & Giller, 2019; Gilles et al., 2014; Mercier & Hohmann, 2002).

In recognizing that 46% of audio engineers are without formal audio engineering education, the majority of survey respondents, along with WHO (2022) and Mulder et al. (2021), one potential "gap" filler could provide education/certification to audio engineers, especially for those operating over $L_{Aeq}85$ dB. Content could include WHO Safe listening recommendations, Mulder et al.'s suggestions on monitoring, sound levels, and the following:

- Recognizing and encouraging appropriate interpersonal behavior among all staff, building on mutual respect that fosters 'proper' relationships,
- Creating a shared vision for all participants to manage expectations,
- Encouraging mentorship,
- Managing staff fatigue,
- Fostering hearing management that promotes hearing longevity and critical listening,
- Actively managing safety versus noise exposure through zones, warnings, provision of earplugs and managing performance programming, dynamics, and participant placement, educating the public on personal safety and responsibility,
- Appropriate management of interactions between attendees and audio engineers, and
- Encouraging education and staff development on matters least understood by most engineers, such as sound level monitoring.

6.4 CSR Ethical Theory and Live Sound

CSR Ethical Theory suggests businesses have a fiduciary duty (involving trust) based on ethical responsibilities to consider and give attention to the legitimate interests of each group of stakeholders, balancing the multiplicity of interests for mutual benefit, justice, cooperation, and the common good (Bowen et al., 2013). Mutual benefit in live sound events provides a financially viable, pleasurable experience that encourages respect between participants with consideration for their safety and well-being.

Each live sound participant has different expectations, motivations, preferences, biases, tolerance, and propensity for safety or tolerance of risk. Summing up this underlying sentiment, one respondent said that the sound might be “...*too quiet and too loud at the same time. Too much reverb, or not enough.*” Making appropriate sound choices that please each participant to maximize event viability could be difficult, especially when most audio engineers said they experienced pressure to perform tasks in ways they would otherwise not prefer. Some engineers weigh the various influences and pressures against their own beliefs and decide accordingly. Some choose to ignore external influences, and some believe they are “...*there to serve, balancing the needs of the musicians and audience and those with authority.*” Alongside the multiplicity of opinions, responsible audio engineers must consider the legitimate interests of participants and the common good by considering their safety and well-being.

As mentioned previously, one facet of live sound that attracts significant differences of opinion is sound levels, where a mix may be appropriately balanced, but the sound level is higher or lower than participants prefer. There is tension between the traditional model of loud events favored by many participants and noise exposure that can threaten hearing longevity. Even governing authorities differ in their noise level guidance. Regulations for workplace safety commonly provide sound-level limits that typically allow greater risk than WHO World Report on Hearing (WHO, 2021). In March 2022, the WHO released a WHO Global Standard for Safe Listening Events and Venues (WHO, 2022) that aims to balance artistic integrity, audience satisfaction, and safety. This 2022 guideline recognizes the challenges events have in meeting safe limits and proposed an acceptable level of risk that, in their 2021 Report, would have only allowed just 15 min of exposure (WHO, 2021).

Breaching fiduciary duty can potentially negate an event’s economic benefit and the ideals of responsible ‘power.’ For example, a 2018/19 UK High Court awarded a viola player £750,000 due to hearing damage caused during a rehearsal at the Royal Opera House (Coleman, 2016–2018). Given this precedent, those who manage sound levels and exposure could potentially be legally responsible for the hearing management of people in their care.

Differences in knowledge and sense of responsibility could feasibly create pressure between people. One person could follow the Government’s recommendations for guidance on noise levels and exposure. Another may follow the 2021 WHO World Report on Hearing (WHO, 2021). Yet another may continue traditional practice, while another could follow the 2022 WHO Global Standard for Safe Listening Venues and Events (WHO, 2022). An event organizer, an audio engineer, a performer, and an attendee could each have different views on sound levels, what an ‘appropriate’ mix is, and who is responsible for managing sound exposure.

Although differing opinions have value, an agreed-upon informed “vision” could then give attention to “legitimate interests,” encourage eustress, and reduce the pressure through a common understanding. Adherence to CSR Ethical Theory would require that those in a position of responsibility weigh up informed participant concerns while also considering participants that may not be informed or capable of making informed choices to “maximize the mutual benefit for all, or at least the majority.”

7 Conclusion

This study confirms that audio engineer choice is affected by the physical and social influences and pressures within the live sound culture. While not always enjoyable, navigating interpersonal interactions is essential in maximizing the event experience and viability. While the various voices should be considered, the decisions that affect people’s health, safety, well-being, and financial investment should be agreed upon by informed and responsible people. A shared vision and a respectful, collaborative approach between participants could reduce negative influences and pressures while also encouraging development and eustress, which helps balance tradition, safety, musicality, art, and individual preference and motivations.

The corporate social responsibility model has been beneficial as a comparative ideal for this study. Actioning many of the themes within the model could provide significant benefits to the live sound culture. Meeting CSR objectives would produce long-term profits, responsibly use power, integrate social demands, and contribute to a good society by doing what is ethically correct.

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Data Availability Due to the survey’s ethical conditions, and the sensitive and personal nature of some of the responses, the raw data is unavailable to other researchers for further analysis.

Declarations

Ethical Approval The international survey was approved by, and with the oversight of, the University of Canterbury Ethics Committee, 5 October 2019, Ref HEC 2019/135.

Informed Consent In the information sheet at the start of the online survey, participants were informed that “The results of the project may be published ... but [the participant’s] identity will not be made public.” By continuing with the survey, participants have provided sufficient consent to publish any non-identifying content. No person could be identified from any of the content used in the article.

Competing or Conflicts of Interest The author declares that they have no competing or conflicts of interest.

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